

Abstract

Structured channel pruning is widely used to compress convolutional networks for practical vision deployment, yet most pruning criteria are driven by local structural cues and do not test whether a channel is semantically redundant across class-specific activation patterns. We present GRA-CNN, a semantic-aware structured pruning framework that combines a structural anchor based on channel independence, Taylor sensitivity, and Fisher information with a quality-gated boundary refinement based on Gray Relational Analysis (GRA). The GRA branch estimates class-aware inter-channel redundancy and is activated only in layers where the semantic signal is sufficiently reliable. As a result, semantic scoring acts as a targeted correction near the keep/prune boundary rather than a full replacement for structural ranking. We evaluate the method on ResNet-56 and VGG-16 on CIFAR-100, a five-ratio transfer sweep on ResNet-18/Tiny-ImageNet, strict equal-budget comparisons, multi-seed component ablations, and device-level latency profiling. The results show a consistent advantage over CHIP in high-compression residual settings on ResNet-56, while VGG-16 shows a more mixed, architecture-dependent pattern. The ablation study shows that channel attention, boundary refinement, and quality gating all contribute, with the largest penalties appearing at aggressive compression. Overall, GRA-CNN is best viewed as a robustness-oriented semantic refinement for structured pruning: it is most useful when structural-only rankings become brittle, rather than a universally dominant score across all architectures and compression levels.